ASSIGNMENT-3

Clustering Analysis Report

Understanding and Implementing K-Means, Hierarchical, and DBSCAN Algorithms

1. Objective

The objective of this assignment is to understand and implement different clustering techniques such as **K-Means**, **Hierarchical Clustering**, and **DBSCAN**. These algorithms help in identifying hidden patterns and grouping similar data points in an unlabeled dataset. The goal is to apply these methods to a real-world dataset and compare their performance.

2. Dataset Description

The dataset used for this analysis is the **East-West Airlines dataset**, which contains information about customer travel patterns, frequent flyer details, and service usage.

The dataset includes features such as:

- Balance
- Qual miles
- Bonus miles
- Flight miles
- Flight transactions
- Days since enrollment

This dataset is useful for identifying customer segments based on their behavior.

3. Data Preprocessing

Before applying clustering algorithms, the dataset was preprocessed as follows:

- Removed unnecessary text rows and irrelevant columns
- Handled missing values by removing incomplete records
- Selected only numerical features for clustering
- Standardized the data using scaling techniques to ensure all variables contribute equally

Data scaling was essential because clustering algorithms are distance-based and sensitive to feature magnitude.

4. Exploratory Data Analysis (EDA)

EDA was performed to understand the structure and distribution of the dataset:

- Histograms were used to analyze the distribution of features
- Box plots helped identify outliers
- Scatter plots were used to visualize relationships between variables

Key observations:

- Some features showed skewed distributions
- Presence of outliers in mileage-related features
- Natural grouping tendencies observed in certain variables

5. Implementation of Clustering Algorithms

5.1 K-Means Clustering

K-Means clustering partitions the dataset into K distinct clusters.

- The **Elbow Method** was used to determine the optimal number of clusters
- Based on the analysis, **K = 3** was selected
- The algorithm grouped customers into clusters based on similarity

Advantages:

- Simple and efficient
- Works well for well-separated clusters

5.2 Hierarchical Clustering

Hierarchical clustering builds a tree-like structure (dendrogram) of clusters.

- Agglomerative approach was used
- Different linkage methods were tested (ward, complete, average)
- Final clusters were formed by cutting the dendrogram

Advantages:

- No need to predefine number of clusters
- Provides hierarchical relationships

5.3 DBSCAN Clustering

DBSCAN groups data based on density.

- Parameters used:
    - Epsilon (eps): Defines neighborhood radius
    - Minimum samples (minPts): Minimum points required to form a cluster
- DBSCAN identified:
    - Core points
    - Border points
    - Noise (outliers)

Advantages:

- Handles noise effectively
- Can detect arbitrarily shaped clusters

6. Cluster Analysis and Interpretation

K-Means Clusters:

- Cluster 1: High-value customers with frequent travel and high mileage
- Cluster 2: Moderate usage customers
- Cluster 3: Low activity customers

Hierarchical Clustering:

- Produced similar groupings to K-Means
- Provided better understanding of cluster hierarchy

DBSCAN:

- Identified dense clusters
- Successfully detected outliers/noise points
- Less effective when clusters have varying densities

7. Visualization

The clustering results were visualized using scatter plots:

- Different colors were used to represent different clusters
- Clear separation was observed in K-Means
- Hierarchical clustering showed structured grouping
- DBSCAN highlighted noise points distinctly

8. Evaluation Metrics

To evaluate clustering performance, **Silhouette Score** was used:

- **K-Means:** Highest score, indicating well-defined clusters

- **Hierarchical Clustering:** Slightly lower but comparable performance
- **DBSCAN:** Lower score due to noise and density variation

Conclusion from evaluation:
K-Means performed best for this dataset.

9. Conclusion

In this assignment, three clustering algorithms were successfully implemented and compared.

- **K-Means** performed best in terms of cluster separation and simplicity
- **Hierarchical Clustering** provided deeper insights into data structure
- **DBSCAN** was useful for identifying outliers but less stable for this dataset

Overall, clustering techniques proved effective in segmenting customers, which can help businesses in targeted marketing and decision-making.

10. References

- Scikit-learn documentation
- Machine Learning textbooks and lecture notes
- Online resources for clustering techniques